EC 571: Digital VLSI Circuit Design

Delay Optimization

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Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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Last lecture

- Last lecture
  - Switches, Inverters, Noise Margins
- Today's lecture
  - Inverter delay optimization
- Reading
  - Chapter 5

Power Dissipation

Instantaneous Power $= V(t) * I(t)$

Peak Power $= V_{DD} * \underset{t \to t+T}{I_{peak}}$

Average Power $= \frac{1}{T} \int_{t}^{t+T} P(t)\, dt$

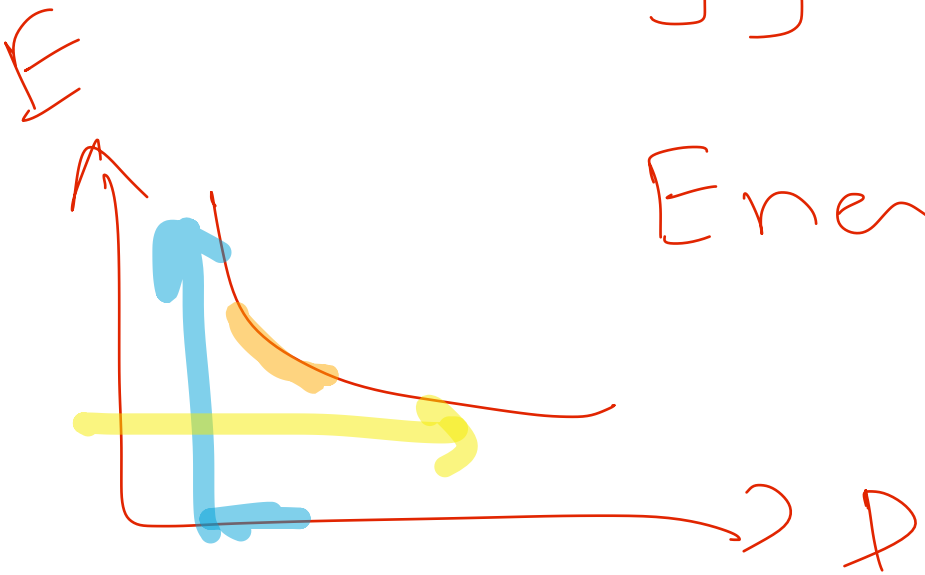
$= \frac{1}{T} \int_{t}^{t+T} V_{DD}\, i(t)\, dt$

$= \frac{V_{DD}}{T} \int_{t}^{t+T} i(t)\, dt$

Power * Delay ?

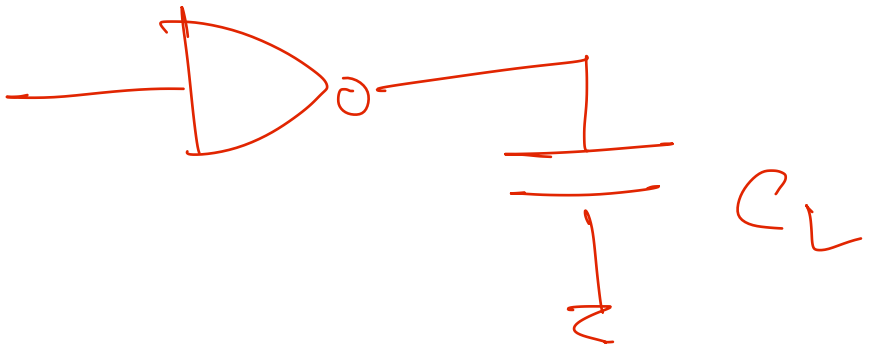
J/s * s ⇒ J

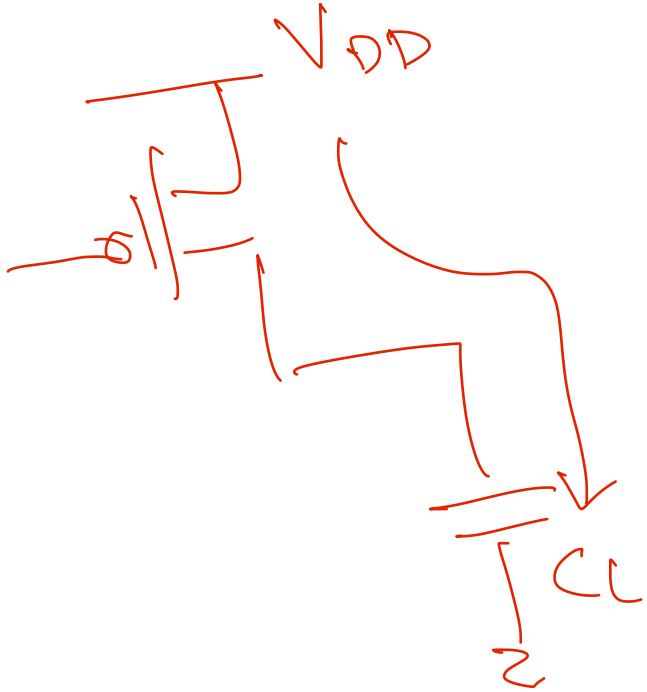
Energy * Delay ? ✓



Energy-efficiency metric

Energy consumption in CMOS ckt





Energy from the power source

$$= \int_0^T V_{DD}\, i_{DD}(t)\, dt$$

$$= \int_0^{V_{DD}} V_{DD} \, C_L \frac{dV_{out}}{dt} \, dt$$

$$= \int_0^{V_{DD}} V_{DD} \, C_L \, dV_{out}$$

$$= C_L V_{DD}^2$$

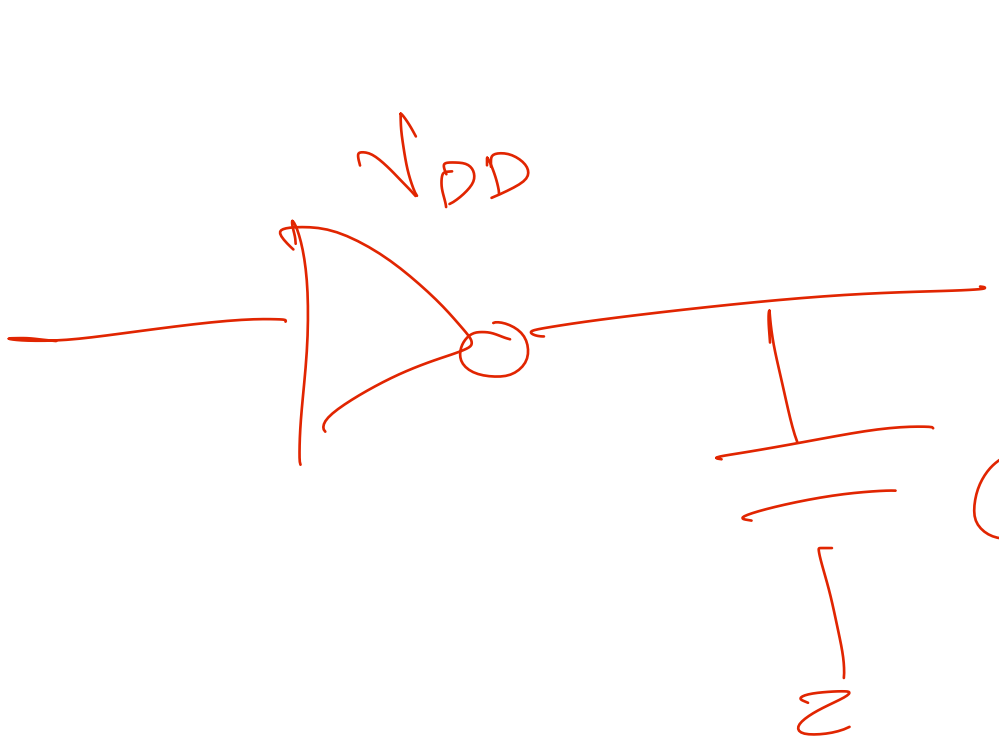
Energy stored in the capacitor

$$E = \int_0^T V_{out}\, i(t)\, dt$$

$$= \int_0^T V_{out} \frac{C\, dV_{out}}{dt} dt$$

$$= \int_0^{V_{DD}} V_{out}\, C\, dV_{out}$$

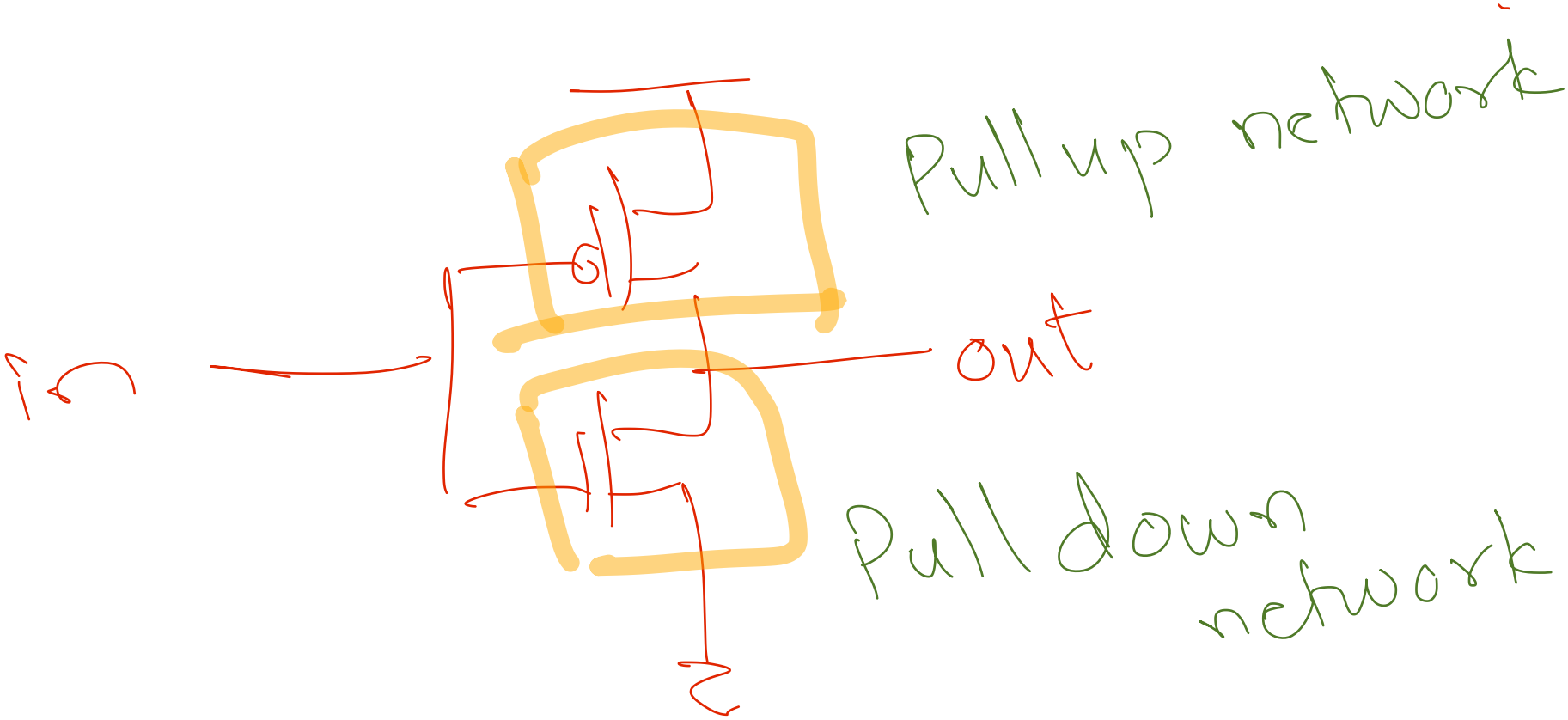
$$= \frac{1}{2} C V_{DD}^2$$

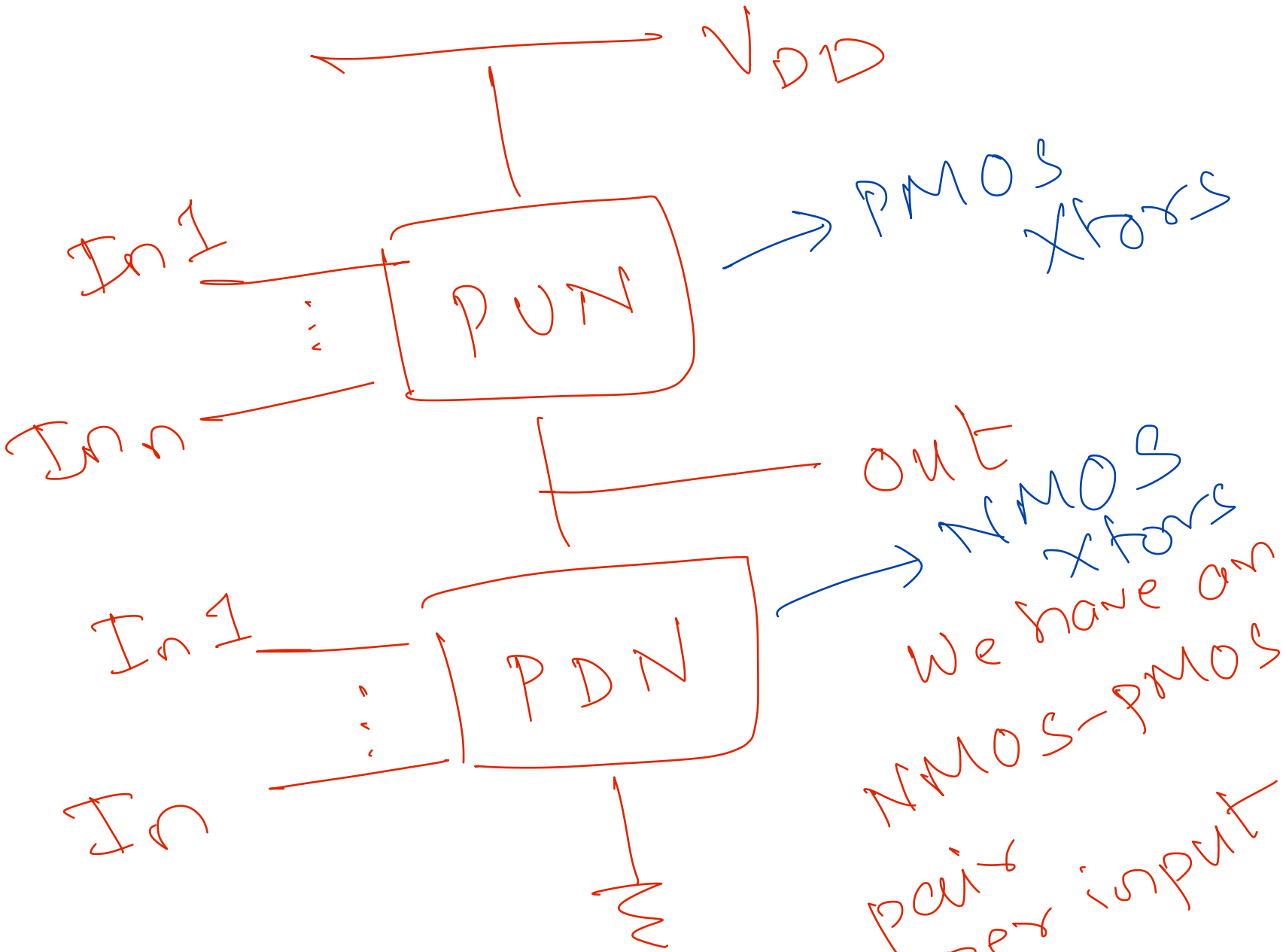


Charged & Discharged 10 times each

$$10 C_L V_{DD}^2$$

Static Complementary CMOS Logic

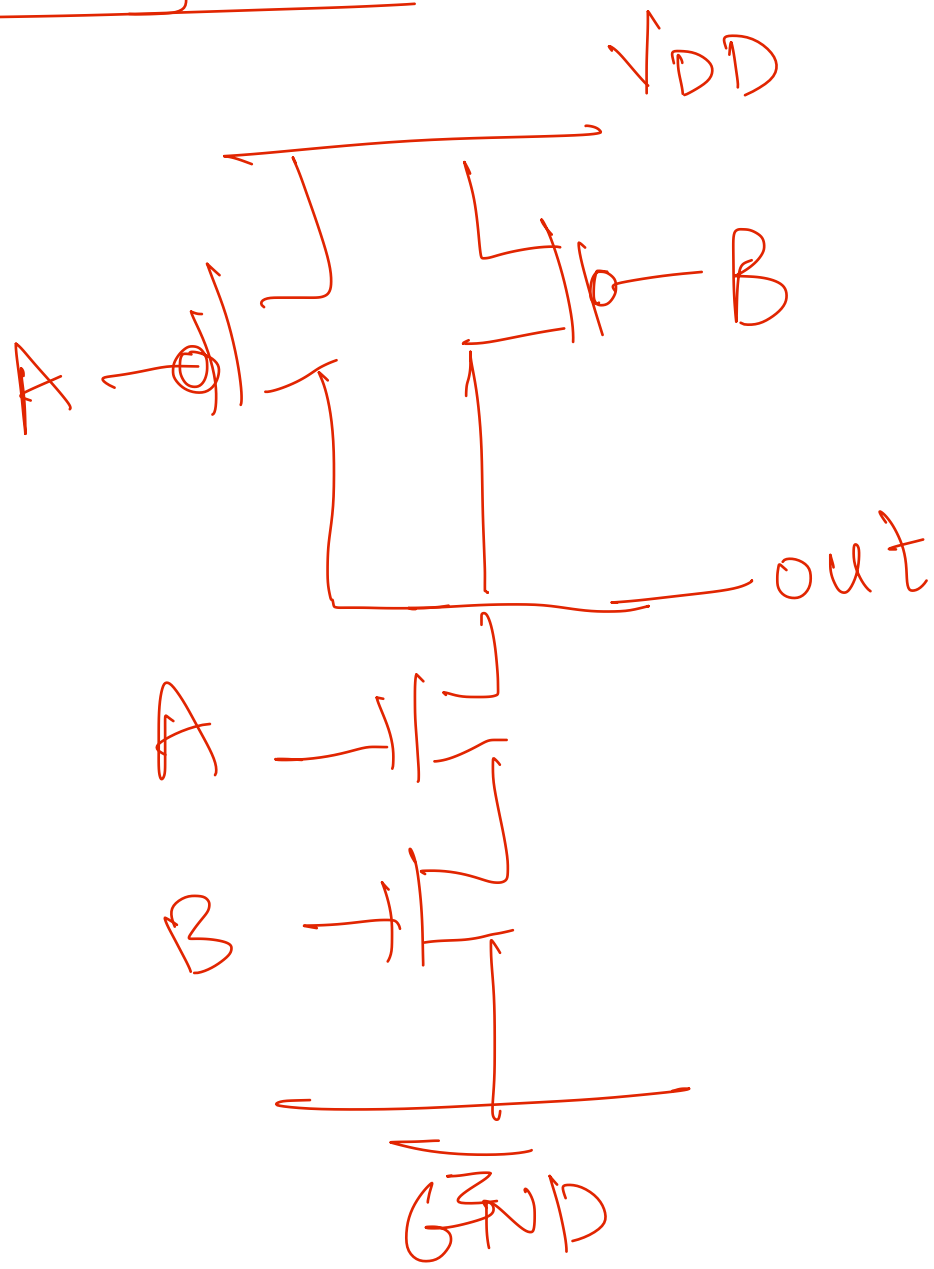




- PUN & PDN are dual logic nlws
- PUN & PDN functions are complementary

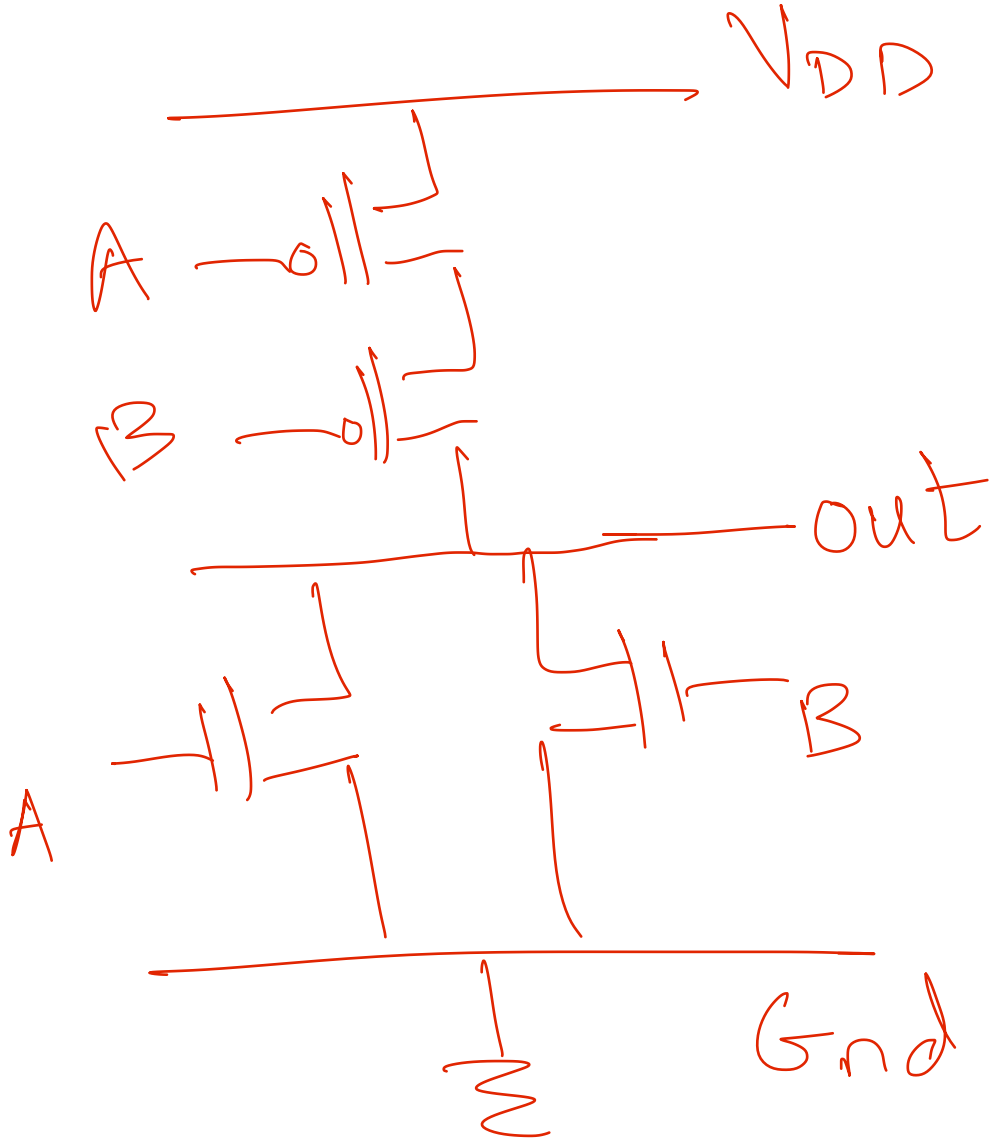
NAND 2 gate using SCL

A	B	out
0	0	1
0	1	1
1	0	1
1	1	0



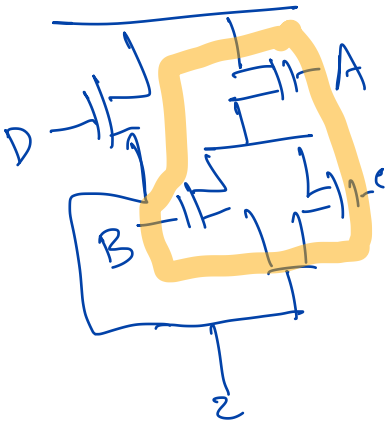
NOR2 gate using SCL

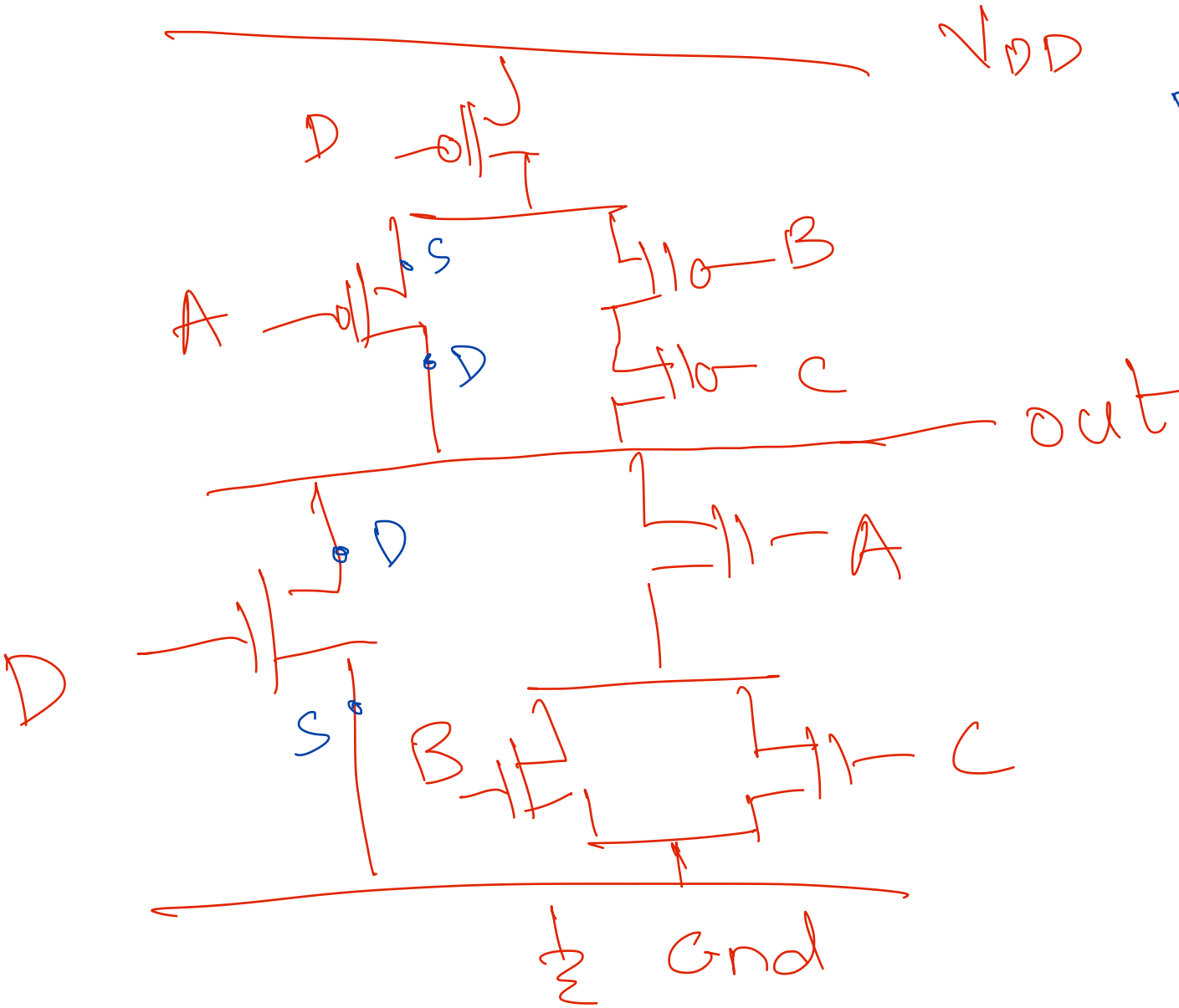
A	B	out
0	0	1
0	1	0
1	0	0
1	1	0



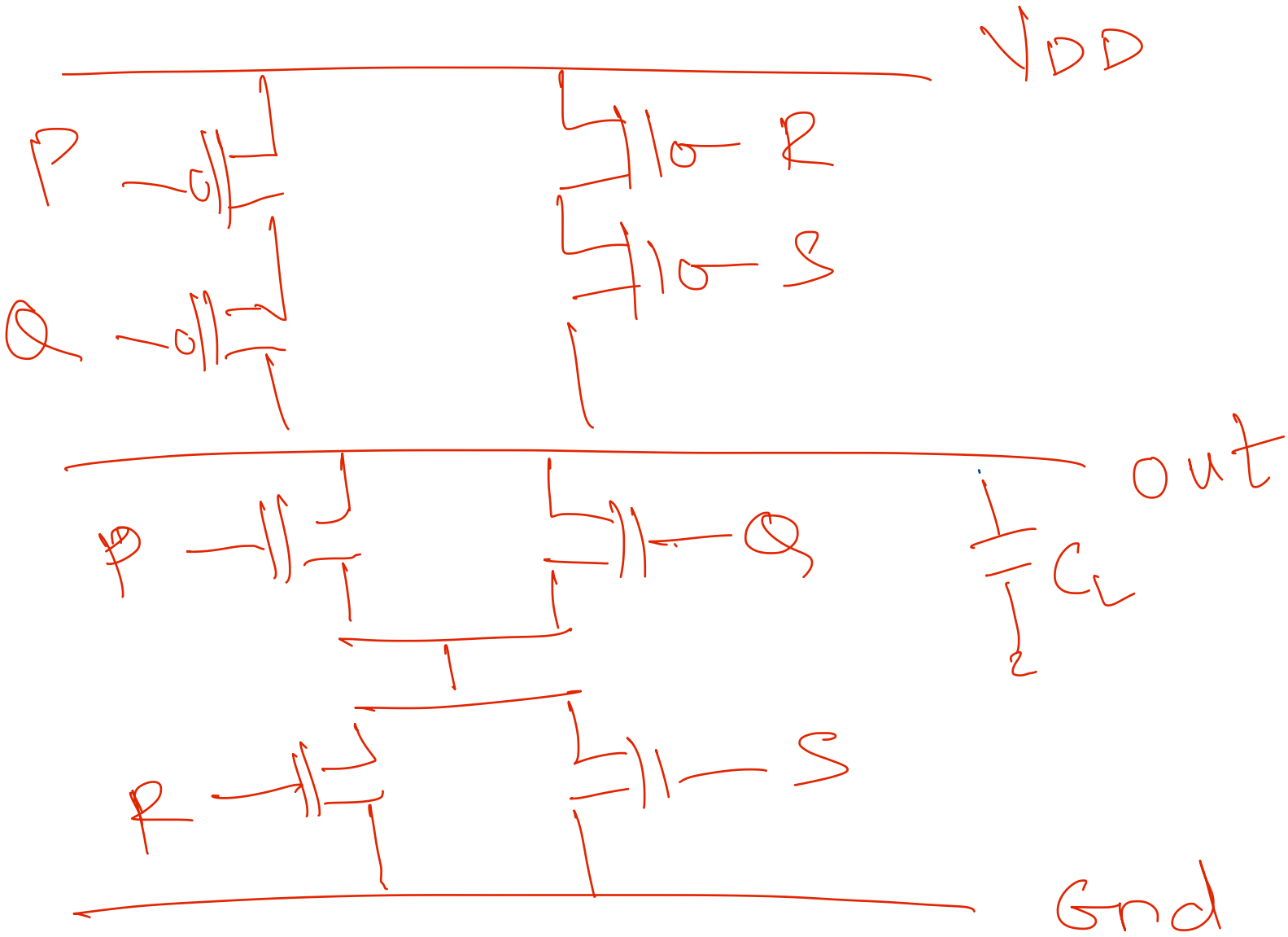
$$F = \overline{D + A \cdot (B + C)} \times$$







$$F = \overline{(P + Q) \cdot (R + S)}$$

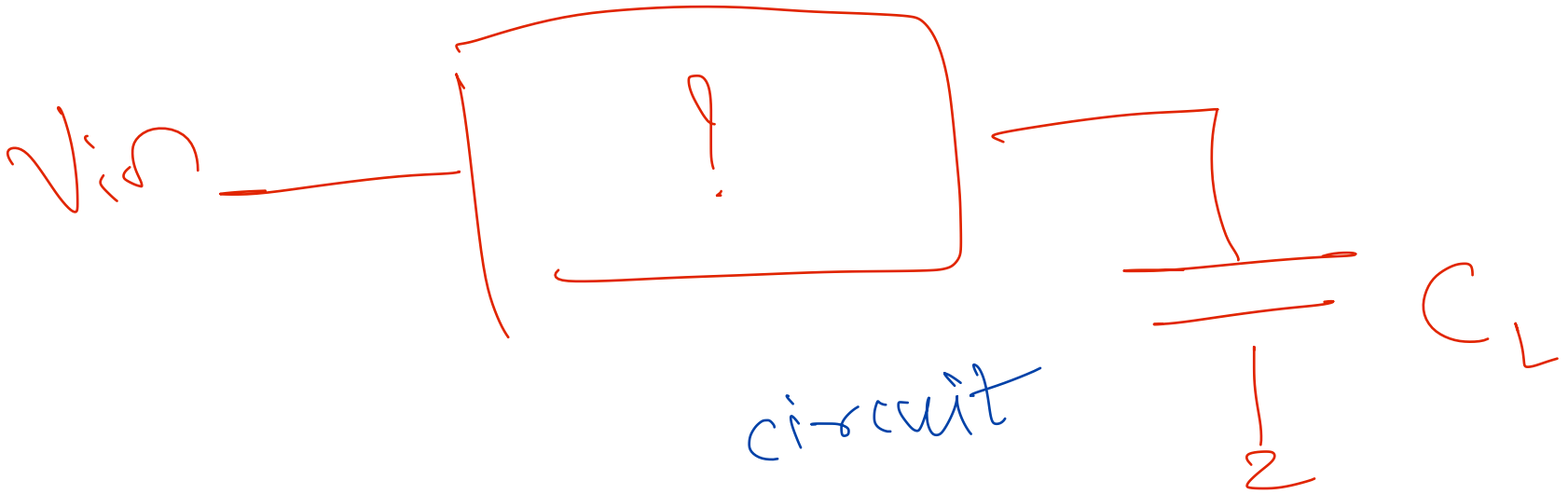


Properties of SCL

- Full rail-to-rail swing at the output
- Symmetric VTC
- No static power dissipation
- Propagation Delay is function of capa & resis of the Xtors

Transistors

Inverter Delay Optimization

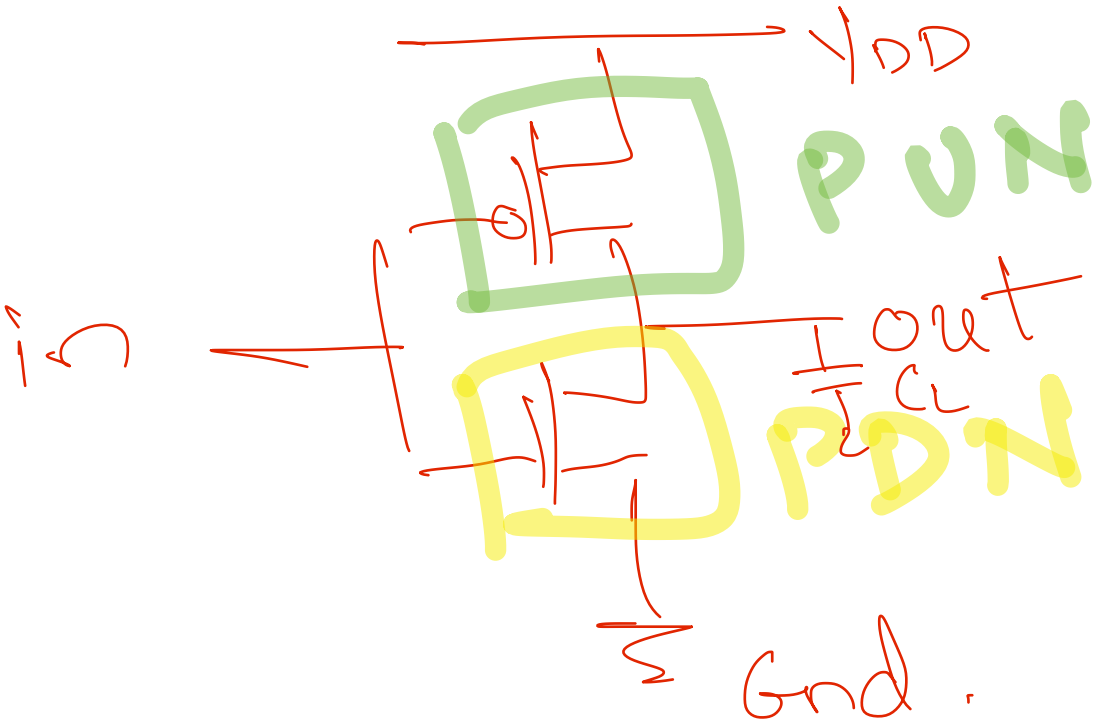


- Design the ckt for driving C_L

O = # inverter stages
Size of the inverters

G = Minimize the delay of the ckt.

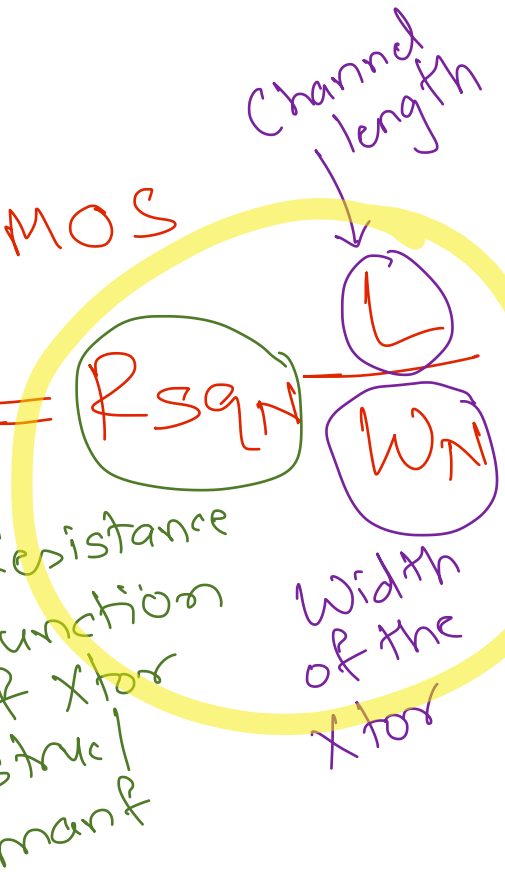
Detour

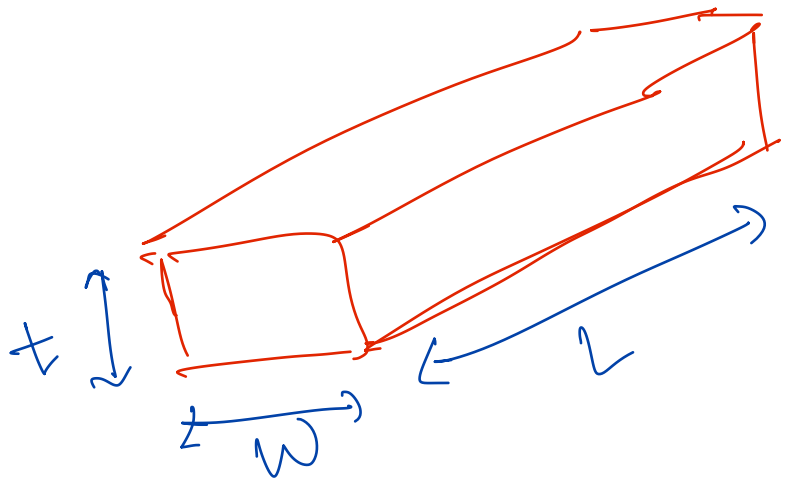


PDN



R_{NMOS}





$$R = \rho \frac{L}{A}$$

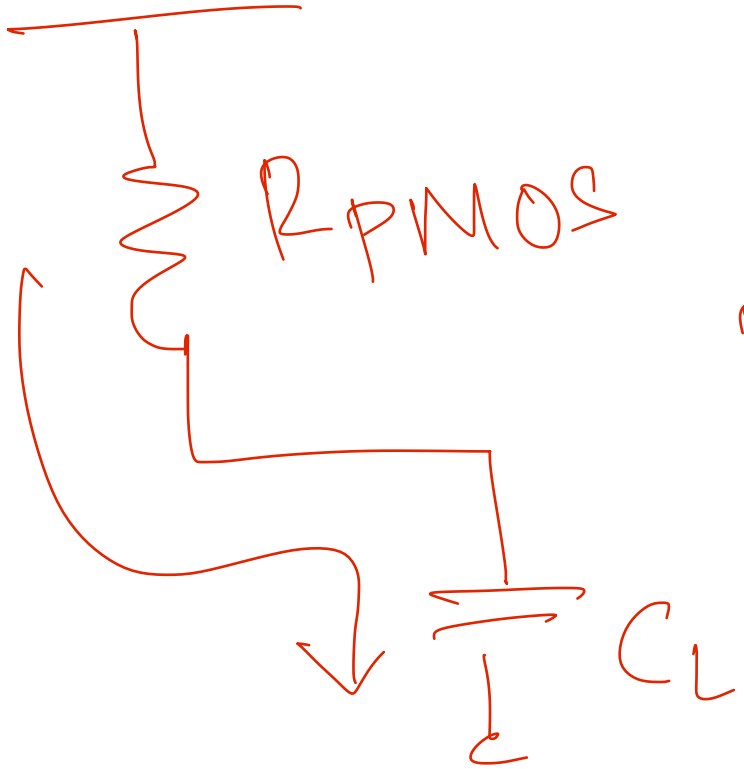
$$R = \rho \frac{L}{t * W}$$

$$= \frac{\rho}{t} \frac{L}{W}$$

R_{SQN} — We can control this

$$\tau_{HL} = R_{NMOS} * C_L$$

$\ln(2)$ → V_{out} → V_{DD} to $0.5 V_{DD}$



$$R_{PMOS} = R_{sqp} \frac{L}{W_p}$$

$$T_{LH} = R_{PMOS} * C_L$$

$$T_{HL} = T_{LH}$$

$$R_{NMOS} \cdot C_L = R_{PMOS} C_L$$

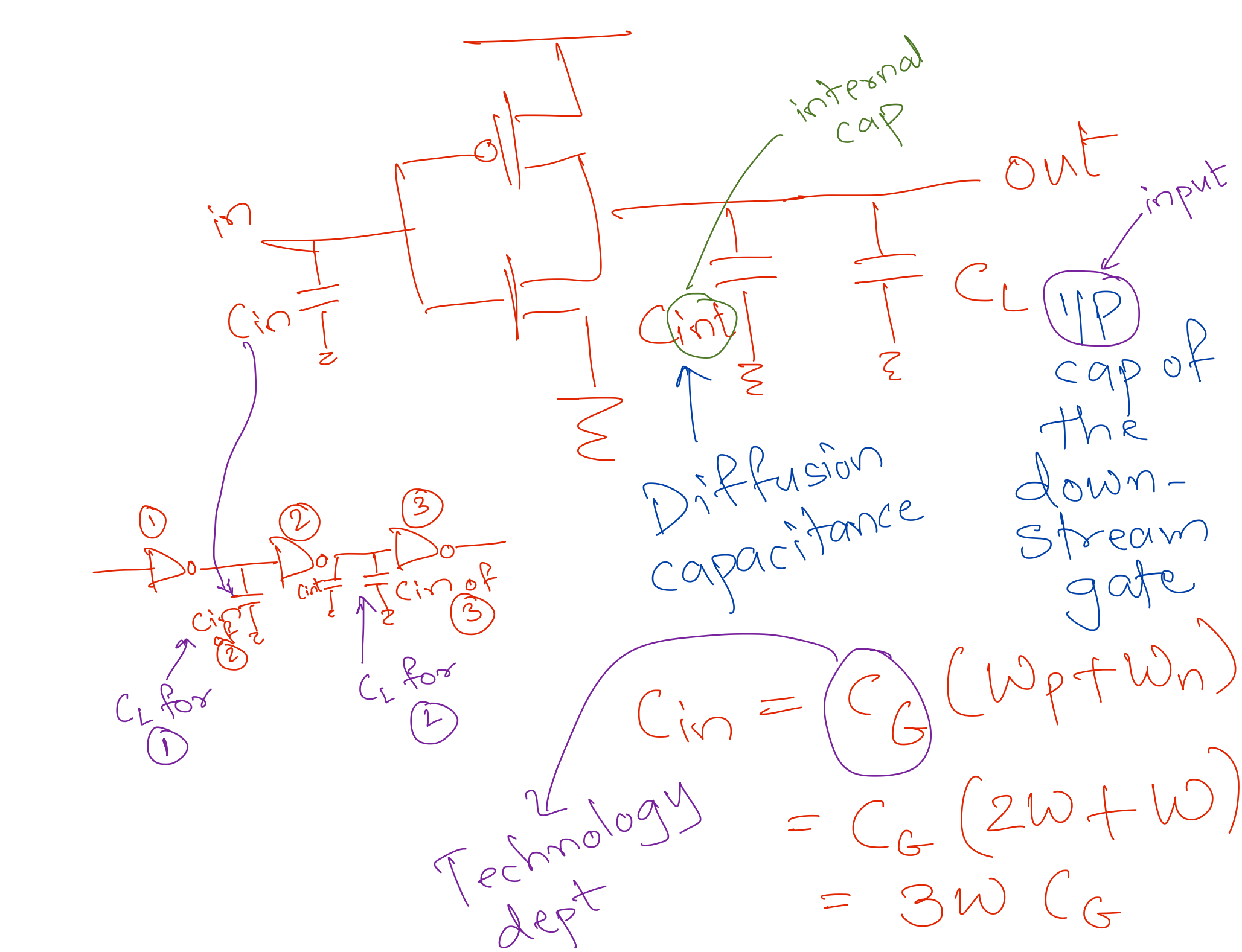
$$R_{NMOS} = R_{PMOS}$$

$$R_{sq_N} \frac{L}{W_N} = R_{sq_P} \frac{L}{W_P}$$

$$R_{sq_P} = \boxed{2} R_{sq_N}$$

Rule of thumb.

$$2 W_N = W_P$$


internal cap
out
in
input
C_{in}
C_{int}
C_L
I/P
cap of the down-stream gate
Diffusion capacitance
C_{in} of ③
C_L for ①
C_L for ②
$C_{in} = C_G (W_p + W_n)$
$= C_G (2W + W)$
$= 3W\, C_G$
Technology dept

$$C_{int} = C_D (W_p + W_n)$$

$$= C_D (2W + W)$$

$$= 3W C_D$$

$$\tau_{HL} = \ln(2) \cdot R_{NMOS} (C_L + C_{int})$$

$$= \ln(2) \cdot R_{NMOS} C_{in} \left(\frac{C_L}{C_{in}} + \frac{C_{int}}{C_{in}} \right)$$

$$= \ln(2) \cdot R_{NMOS} C_{in} (f + \gamma)$$

$$\gamma = \frac{C_{int}}{C_{in}} = \frac{3 C_D W}{3 C_G W} = \frac{C_D}{C_G}$$

1 to 1.5

Agnostic of xtor sizing

$$f = \frac{C_L}{C_{in}} \leftarrow \text{Electrical fanout.}$$

$$T_{HL} = \ln(2) R_{NMOS} \cdot C_{in} (\gamma + f)$$

$$= \ln(2) R_{sqr} \frac{L}{W_n} C_{in} (\gamma + f)$$

$$= \ln(2) R_{sqn} \frac{L}{W} 3W C_g (\gamma + f)$$

$$= \ln(2) R_{sqn} L \cdot 3 C_g (\gamma + f)$$

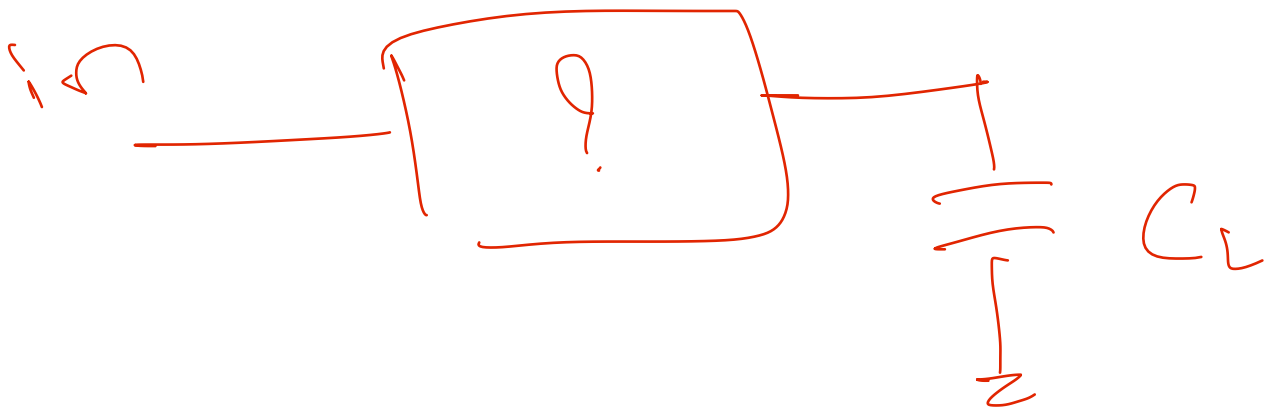
$$= 3 \cdot \ln(2) R_{sqn} L \cdot C_g (\gamma + f)$$

$$= t_{inv} (\gamma + f)$$

Doesn't mean delay of inverter

Design problem

1

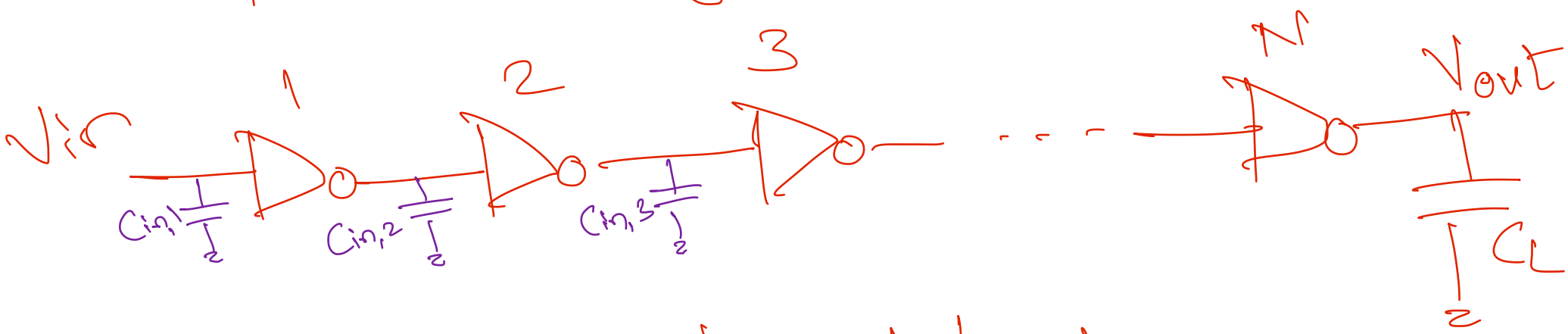


- Size of first inverter
- Load cap
- # inverters

} Known info / Constraints

Minimize delay of inverter chain } Objective

Determine the sizes of the Xtors in our design



$$t_p = t_{p_1} + t_{p_2} + t_{p_3} + \cdots + t_{p_N}$$

$$t_{p_j} = t_{inv}(\gamma + f) = t_{inv}\left(\gamma + \frac{C_{in,j+1}}{C_{in,j}}\right)$$

$$t_p = \sum_{i=1}^{N} t_{pj}$$

$$= t_{inv} \sum_{i=1}^{N} \left(\gamma + \frac{C_{in,j+1}}{C_{in,j}} \right)$$

- `N-1` unknowns
- To minimize delay take `N-1` partial derivatives

$$\frac{\partial t_p}{\partial C_{in,j}} = t_{inv} \frac{1}{C_{in,j-1}} - t_{inv} \frac{C_{in,j+1}}{C_{in,j}^2}$$

$$\frac{\partial t_p}{\partial C_{in,j}} = 0 \Rightarrow$$

$$t_{inv}\frac{1}{C_{in,j-1}} - t_{in}\frac{C_{in,j+1}}{C_{in,j}^2} = 0$$

$$\frac{C_{in,j}}{C_{in,j-1}} = \frac{C_{in,j+1}}{C_{in,j}} = f$$

Electrical fanout

$$\frac{C_{in,2}}{C_{in,1}} = \frac{C_{in,3}}{C_{in,2}} = \cdots = \frac{C_L}{C_{in,N}} = f$$

$$\frac{C_{in,2}}{C_{in,1}} * \frac{C_{in,3}}{C_{in,2}} * \cdots * \frac{C_L}{C_{in,N}} = f^N$$

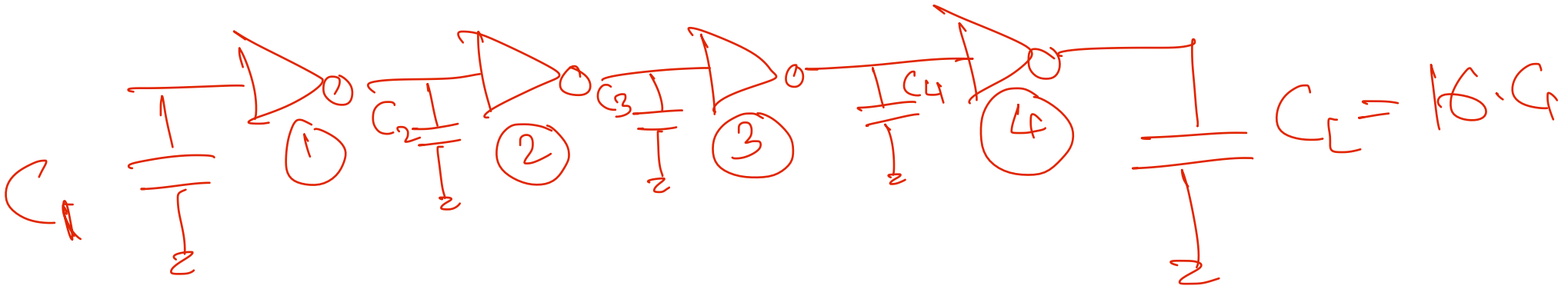
$$\frac{C_L}{C_{in,1}} = f^N = F$$

$$f = \sqrt[N]{F}$$

$$t_{one\ inverter} = t_{inv}(\gamma + f)$$

$$t_p = N \cdot t_{inv}(\gamma + N\sqrt{F})$$

Example



$N = 4$

C_1 gives the size of the Xtors in (1)

$$C_1 = (W_n + W_p)\, C_G$$

$$= (W + 2W)\, C_G$$

$$= 3W \cdot C_G$$

$$f = N \sqrt{\frac{C_L}{C_1}}$$

$$C_G(W_{p_2} + W_{n_2}) = 4 \sqrt{\frac{16 C_1}{C_1}} = 2$$

$$\frac{C_2}{C_1} = \frac{C_3}{C_2} = \frac{C_4}{C_3} = \frac{C_L}{C_4} = 2$$

$$C_G(W_{p_1} + W_{n_1})$$

Optimization problem 2

- Size of first inverter } Constraints
- Size of load cap

- Minimize Delay → Objective.

- Determine # inverter & size of inverters

$$t_p = N\, t_{inv}\left(\gamma + N\sqrt{\frac{C_L}{C_{in}}}\right) \quad \text{--- (1)}$$

$$f^N = \frac{C_L}{C_{in}} \qquad N = \frac{\ln(C_L/C_{in})}{\ln(f)} \quad \text{--- (2)}$$

Using (1) and (2)

$$t_p = \frac{\ln(C_L/C_{in})}{\ln(f)} \cdot t_{inv}(f+\gamma)$$

$$\frac{\partial t_p}{\partial f} = t_{inv} \ln\left(\frac{C_L}{C_{in}}\right) \cdot \left(\frac{\ln(f) - 1 - \gamma/f}{\ln^2 f}\right)$$

$$\frac{\partial t_p}{\partial f} = 0 \quad \text{to minimize delay}$$

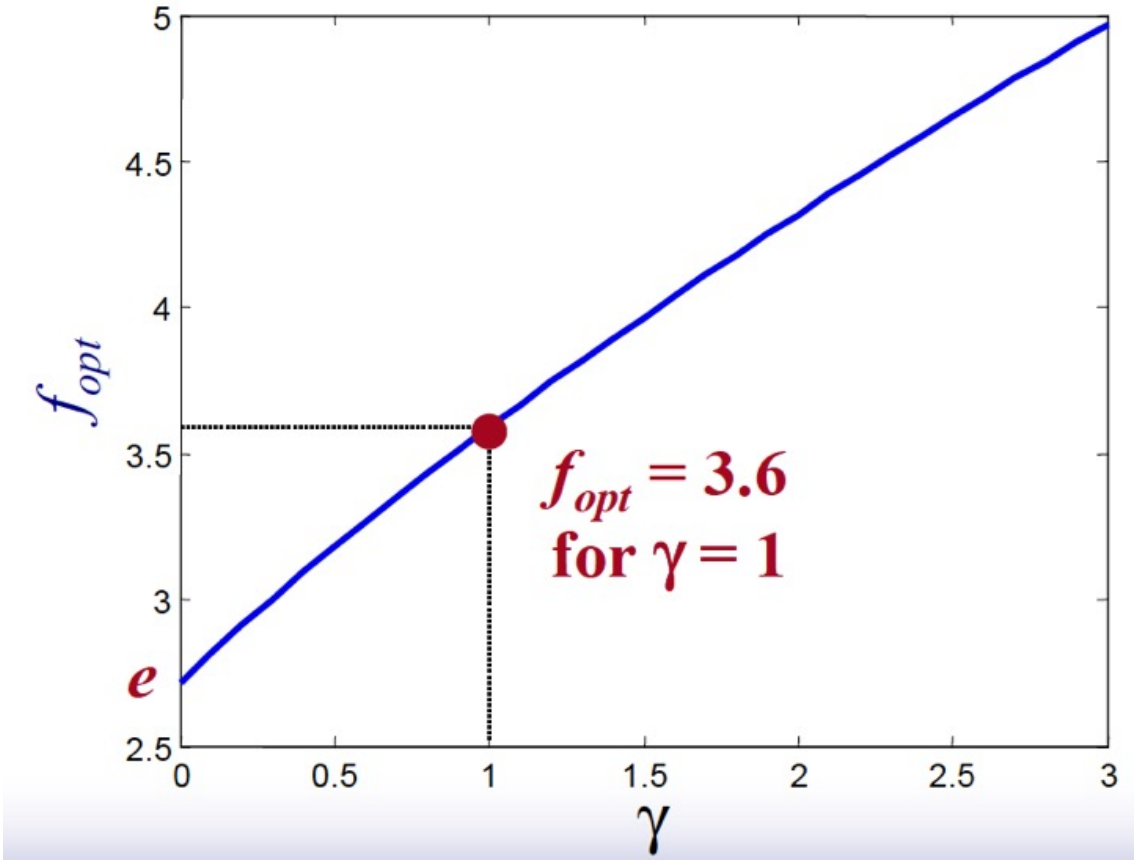
$$f = \exp\left(1 + \gamma / f\right)$$

$$\text{If } \gamma = 0 \Rightarrow f = e$$

$$N = \ln\left(\frac{C_T}{C_{in}}\right)$$

Optimum effective fanout f

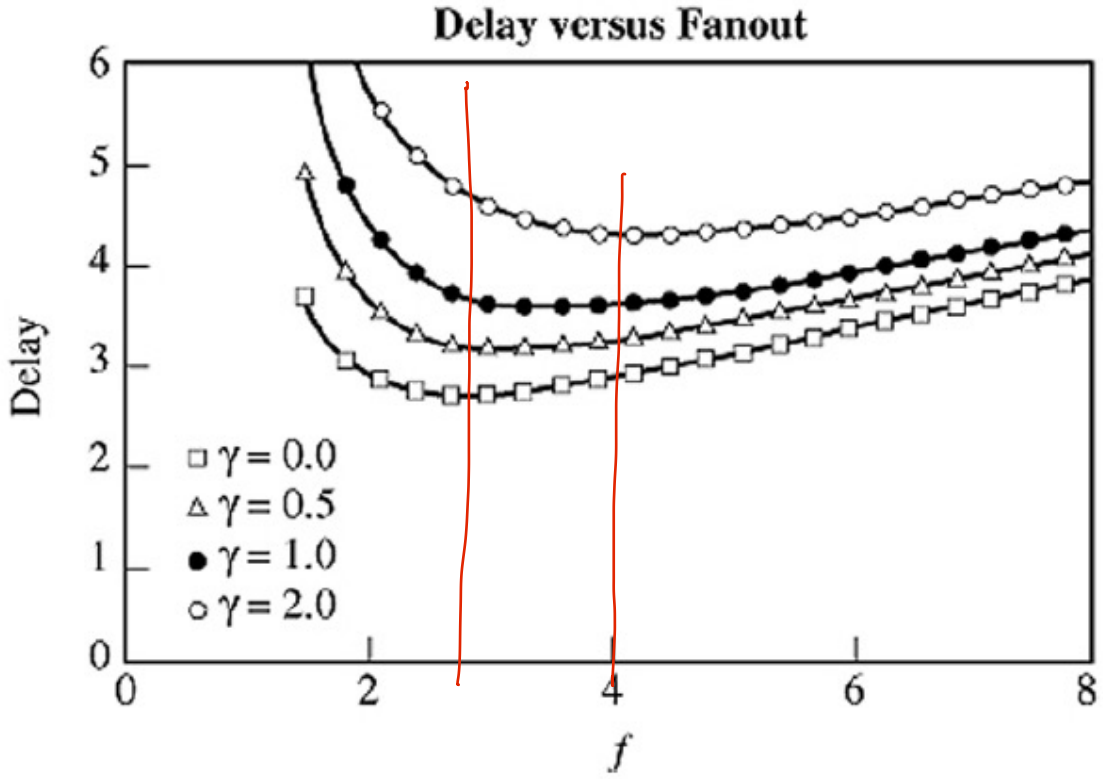
- Optimum f for a given process defined by γ



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In Practice: Plot of Total Delay

- Curves very flat for f > 2
  - Common rule-of-thumb f = 4



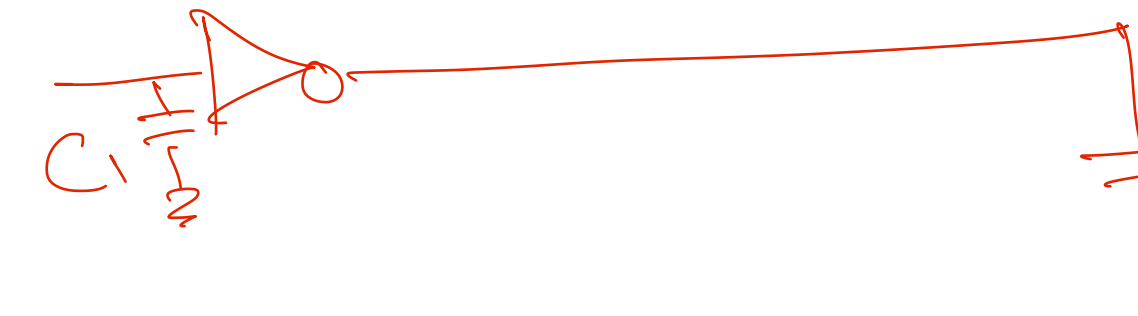
[Hodges, p.281]

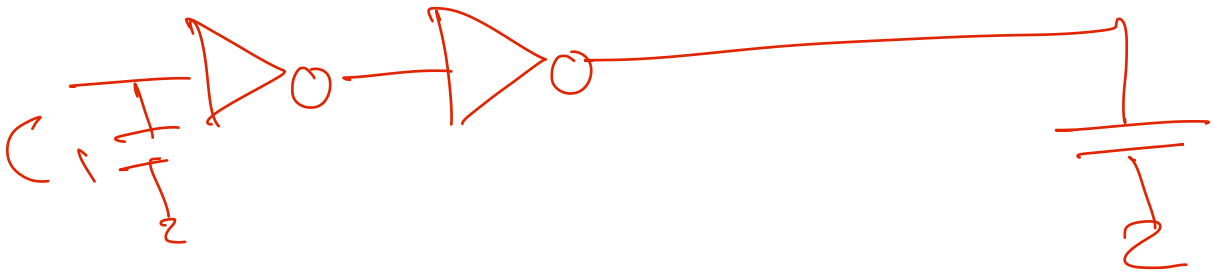
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Example:

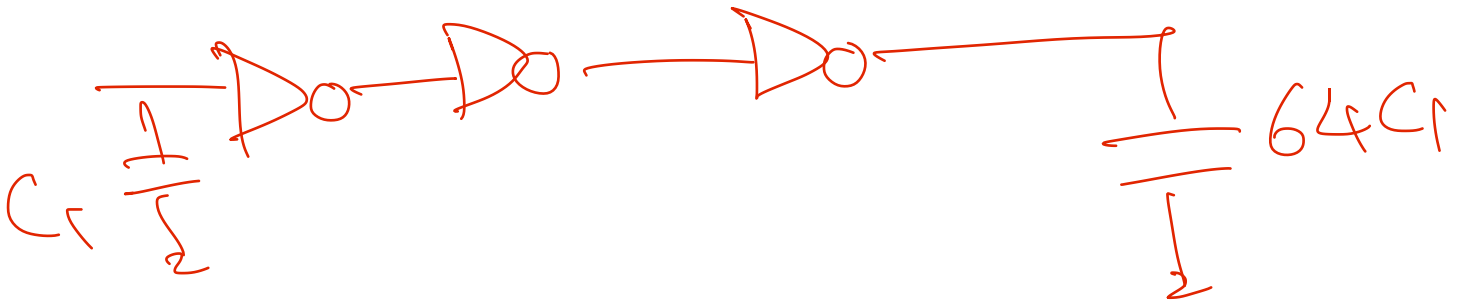
$$t_p = N\, t_{inv} \left(\gamma + N\sqrt{F}\right)$$

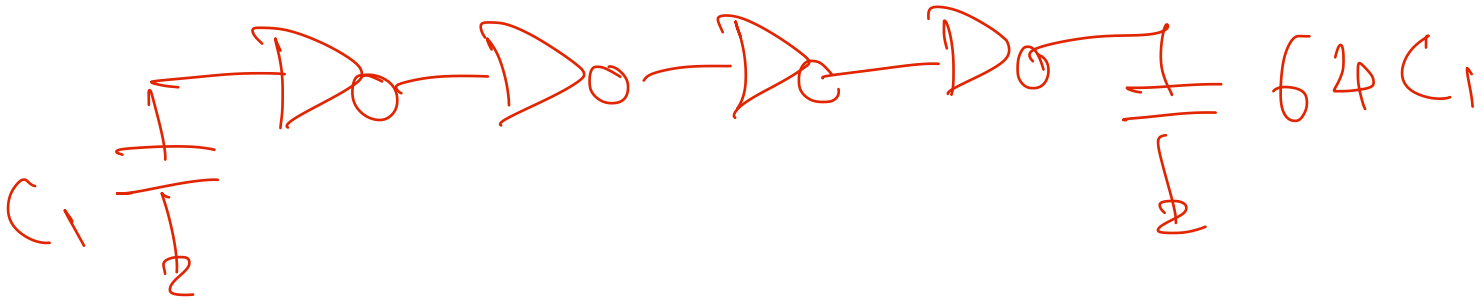
$$\gamma = 1 \qquad F = \frac{C_L}{C_{in}}$$





N	f	t_p
1	64	$65\,t_{inv}$
2	8	$18\,t_{inv}$

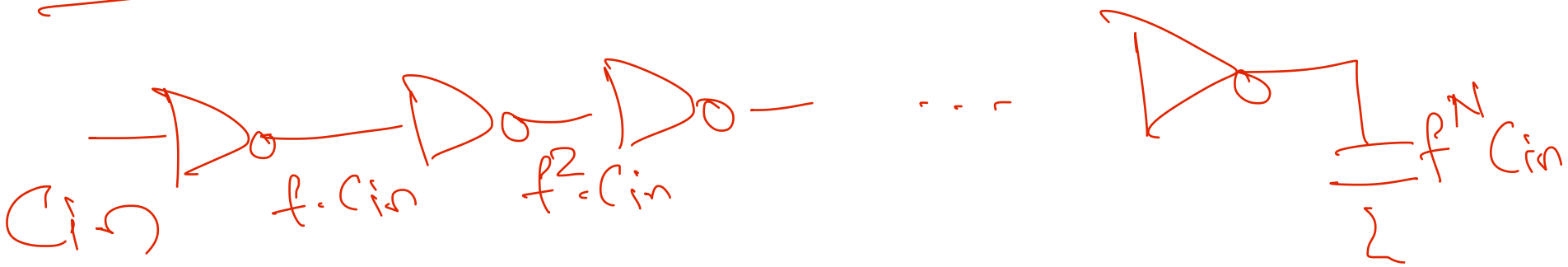




3 2.4 15tinv

4 2.8 15.3tinv

Overhead in energy and area



$$C_{tot} = C_{in} + f C_{in} + f^2 C_{in} + \cdots + f^N C_{in}$$

$$= C_{in}\left(1 + f + f^2 + \cdots + f^N\right)$$

$$= C_{in}\left(1 + f^N\right) + \underbrace{C_{in}\left(f + f^2 \cdots + f^{N-1}\right)}_{\text{overhead}}$$

$C_L = 20pF$, $C_{in} = 50ff$

$\Rightarrow N = 6$ (from prior design approach)

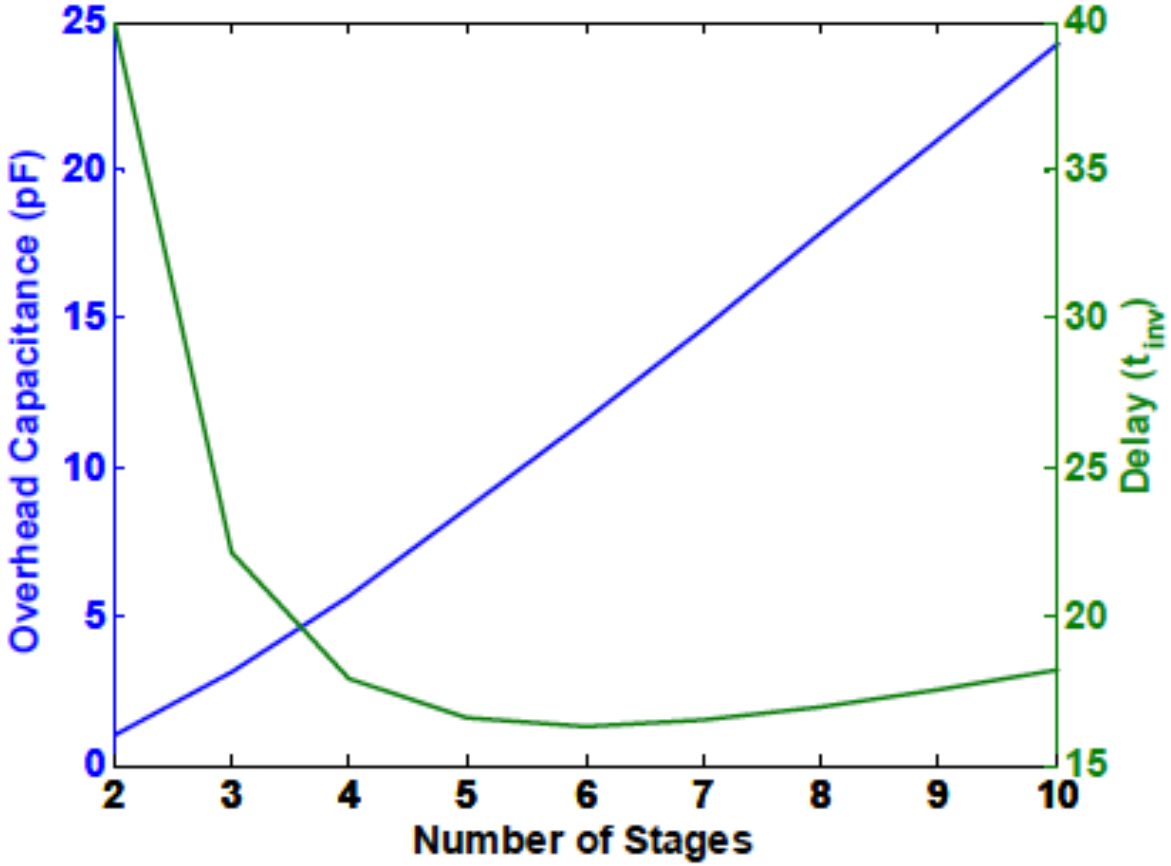
Fixed $= 20pF + 50fF$

Overhead $= 11.6\ pF$

$\Rightarrow$ 50% overhead

Example

Example: C_L = 20pF; C_{in} = 50fF



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